PROCESS MODELING OF AMMONIA VOLATILIZATION FROM AMMONIUM SOLUTION AND MANURE SURFACES: A REVIEW WITH RECOMMENDED MODELS

F. Montes, C. A. Rotz, H. Chaoui

ABSTRACT. *Ammonia emissions from animal feeding operations have become a concern due to their potential adverse effects on animal and human health and the environment. Emissions occur from manure surfaces on the barn floor, during storage, and following field application. Process-based modeling with computer simulation is a cost-effective procedure for quantifying and evaluating emissions across diverse production systems. Based upon theoretical principles and associated published information on ammonia emissions, relationships were refined for modeling the dissociation constant, Henry's law constant, and the mass transfer coefficient to improve predictions of ammonia loss from manure surfaces. Inconsistencies found in widely used expressions for the dissociation constant and mass transfer coefficients in ammonia volatilization models from manures are discussed. Refined expressions were developed that relate these parameters to the temperature, pH, and ionic strength of the ammonium-containing material, and the velocity of air flowing over the material. These expressions were tested by comparing predicted ammonia emission rates to values measured in controlled laboratory experiments for buffered ammonium-water solutions and dairy cattle manure. Experimental results compared well to values predicted using these theoretical expressions derived from ammonia volatilization literature. These process-based relationships provide a basis for developing predictive tools that quantify management effects on ammonia emissions from farms and thus assist in the development and evaluation of strategies for reducing emissions.*

Keywords. *Ammonia, Dissociation constant, Emission model, Henry's law constant, Manure, Mass transfer coefficient.*

Support for the Clean Air Act has increased pressure on regulatory agencies to address airborne emissions from animal feeding operations (NRC, 2003). Ammonia (NH_3) emitted in livestock production is of particular interest because of the potential environmental impact and the loss of nitrogen (N), a valuable nutrient that needs to be replaced with costly petroleum-based fertilizers. Ammonia in the atmosphere can precipitate in acid rain, contributing to surface water eutrophication and over-fertilization of ecosystems, and it can contribute to the development of small particulate matter, which is a human health concern (Arogo et al., 2003a; Meng et al., 1997; Renard et al., 2004; Van Breemen et al., 1982).

Determining the amount of NH_3 lost from animal feeding operations is difficult. Accurate measurement of NH_3 volatilization is expensive, and measurements obtained from a particular animal production facility cannot be generalized to other operations because emissions are affected by differences in production conditions, environment, and management (Arogo et al., 2003a; Arogo et al., 2006; Harper, 2005; Ni and Heber, 2001). Furthermore, accurate ventilation air exchange rates in animal facilities, an essential component in the determination of NH_3 emissions, are difficult to measure. The National Research Council's Ad Hoc Committee on Air Emissions from Animal Feeding Operations identified the limitation of available methodologies to estimate national emissions from animal agriculture and recommended that the U.S. EPA develop a process-based modeling approach (NRC, 2003). Mechanistic models representing the NH_3 volatilization processes as influenced by production and environmental conditions provide robust tools for evaluating management influences on emissions (Arogo et al., 2006; Ni, 1999). Although many models have been used to predict NH_3 emissions, relationships representing dissociation and mass transfer vary considerably among models (Ni, 1999).

The objective of this work was to review the fundamentals supporting NH_3 volatilization models and suggest steps for reducing error and improving consistency and accuracy in the prediction of emissions from manure sources in dairy cattle production systems. An improved understanding of the processes involved in the formation and emission of NH_3 will lead to improved models for evaluating management influences on emissions and developing reduction strategies for all livestock systems.

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PROCESS-BASED MODELS OF AMMMONIA EMISSION

Ammonia volatilization has been correctly described as a mass transfer process, and the most important physical and chemical components are well understood (Arogo et al., 2006; Ni, 1999). Numerous models of this process have been applied in evaluating swine lagoons (Aneja et al., 2001; De Visscher et al., 2001; Koelliker and Miner, 1973; Liang et al., 2002), swine slurry and swine buildings (Anderson et al., 1987; Cumby et al., 1995; Muck and Steenhuis, 1982; Ni et al., 2000; Olesen and Sommer, 1993; Zhang et al., 1994), dairy cow barns (Elzing and Monteny, 1997a, 1997b; Monteny et al., 1998; Muck and Richards, 1983; Muck and Steenhuis, 1980; Webb and Misselbrook, 2004; Zhang et al., 2005), chicken slurries (Hashimoto, 1972; Hashimoto and Ludington, 1971), land application of manures (Génermont and Cellier, 1997; Roelle and Aneja, 2005; Svensson and Ferm, 1993; Van der Molen et al., 1990), and whole-farm production systems (Hutchings et al., 1996; Pinder et al., 2004; Rotz and Oenema, 2006). Although many model applications have been implemented, further evaluation and improvement in predicting emissions from livestock manure are needed (Arogo et al., 2006).

MODEL STRUCTURE

A review of the models listed above and the comprehensive review provided by Ni (1999) suggest a common structure for all volatilization models. At the core of this structure is a convective mass transfer model where the flux varies with ambient temperature, airflow over the surface, and the NH_3 concentration gradient between the ammonium (NH_4^+) solution and the ambient air:

$$J_{NH_3} = h_m\left([NH_3]_{gas} - [NH_3]_{ambient}\right) \quad (1)$$

where J_{NH_3} is the NH_3 volatilization flux (g NH_3-N m^{-2} s^{-1}), h_m is the convective mass transfer coefficient (m s^{-1}), $[NH_3]_{gas}$ is the concentration of gaseous NH_3 in equilibrium with dissolved NH_3 in solution (g NH_3-N m^{-3} air), and $[NH_3]_{ambient}$ is the concentration of NH_3 in ambient air (g NH_3-N m^{-3} air), which is normally very small and assumed to be negligible. The concentration of gaseous NH_3 at equilibrium with the dissolved NH_3 in solution is determined using Henry's law. The Henry's law constant relates the concentration of dissolved NH_3 in water or manure to an equilibrium concentration of NH_3 in the air space immediately above the liquid surface. In engineering terms, the Henry's law constant represents the partition between air and water for a chemical compound in a dilute solution:

$$[NH_3]_{solution} \xleftrightarrow{K_h} [NH_3]_{gas} \quad (2)$$

or

$$K_h = \frac{[NH_3]_{gas}}{[NH_3]_{solution}} \quad (3)$$

where K_h is the Henry's law constant, (g NH_3-N m^{-3} air) / (g NH_3-N m^{-3} solution).

Ammoniacal N in a solution such as manure consists of NH_3-N and NH_4^+-N. The fraction of total ammoniacal N present as NH_3 can be estimated using equilibrium thermodynamic principles:

$$f_{NH_3} = \frac{[NH_3]_{solution}}{\left([NH_3]_{solution} + [NH_4^+]_{solution}\right)} = \frac{1}{1+\frac{[H^+]}{K_a}} = \frac{1}{1+\frac{10^{-pH}}{K_a}} \quad (4)$$

where K_a is the dissociation constant, $[H^+]$ is the proton concentration in solution, pH = -log$[H^+]$, and f_{NH_3} is the NH_3-N fraction of the total ammoniacal N of the solution. Equation 4 assumes that the activity of NH_3 and NH_4^+ is equal to the concentration of these species, and the activity of H^+ is equal to 10^{-pH} where pH is the measured pH of the material. These assumptions are valid for materials composed of dilute aqueous solutions (Morel and Hering, 1993; Snoeyink and Jenkins, 1980; Stumm and Morgan, 1996).

Parameters K_a, K_h, and h_m are important in NH_3 volatilization models, but there is great variability in reported values, and relatively few studies have been committed to verification of these values, particularly in livestock manure systems. The NH_3 dissociation constant used in manure models has typically been estimated by multiplying K_a by an empirical factor that incorporates the effects of ions and solids present in manures. This factor has varied from 0.2 to 1 (Arogo et al., 2003b; Liang et al., 2002; Hashimoto and Ludington, 1971; Van der Molen et al., 1990; Zhang et al., 1994). The variability of values reported for the Henry's law constant of NH_3 has been recognized by researchers in different disciplines (Ni, 1999; Renard et al., 2004; Sander, 1999; Staudinger and Roberts, 1996). In addition, the range of values found in the literature for the mass transfer coefficient extends over three orders of magnitude (Ni, 1999).

RELATIONSHIP AMONG PARAMETERS

Parameters K_a, K_h, and h_m are difficult to isolate in experimental measurements. Measurement of the mass transfer coefficient requires determination of the NH_3 gradient ($[NH_3]_{gas}$ - $[NH_3]_{ambient}$), which in turn requires determination of the NH_3 gas in equilibrium with NH_3 in the solution, and thus knowledge about K_h and K_a. Values for the dissociation constant and Henry's law constant in manure have often been taken from the report by Hashimoto and Ludington (1971). The methodology employed by Hashimoto and Ludington (1971), Zhang (1992), and Arogo et al. (2003b) was based on the observation that when varying the pH of manure while maintaining constant temperature, NH_4^+-N concentration, and the physical conditions for the mass transfer coefficient, the following relationship was valid:

$$\frac{\left(J_{NH_3-N}\right)_{pH}}{\left(J_{NH_3-N}\right)_{pH=12.5}} = \frac{\left(f_{NH_3}\right)_{pH}}{\left(f_{NH_3}\right)_{pH=12.5}} \quad (5)$$

Hashimoto and Ludington (1971) assumed that at a pH of 12.5, the fraction of ammoniacal N present as NH_4^+ was negligible, and therefore $\left(f_{NH_3}\right)_{pH=12.5} = 1$. With this assumption, equation 5 reduces to:

$$\frac{(J_{NH_3-N})_{pH}}{(J_{NH_3-N})_{pH=12.5}} = (f_{NH_3})_{pH} \quad (6)$$

and incorporating equation 4:

$$K_a = 10^{-pH} \frac{(f_{NH_3})_{pH}}{1-(f_{NH_3})_{pH}} \quad (7)$$

Important concerns for using this methodology to determine K_a are: (1) the pH values used are out of the range of those typically found in manure, (2) the implicit assumption that the change in the ratio of species at different pH values is linear, (3) the high rate of volatilization observed when manure or NH_4^+ solution is at a pH of 12.5 (Hashimoto, 1972), and (4) the fact that determining the dissociation constant by the ratio of two volatilization rates includes the compounded variability of h_m and K_h in the final estimate of K_a. Other frequently cited values for K_h and K_a are those of Anderson et al. (1987), which correspond to values from the *Chemical Engineering Handbook* (Perry and Chilton, 1973), or those of Click and Reed (1975), which correspond to values from *Lange's Handbook of Chemistry* (Dean, 1973).

The interconnection between parameters used in alternative models reflects the difficulty in obtaining experimental values of these parameters. When one parameter is experimentally measured, assumed values for the other two add uncertainty to the measured parameter.

The review by Ni (1999) provides a summary of methods that have been used to estimate these parameters and the way in which they have been applied in NH_3 volatilization models. In our work, a quantitative comparison was used to contrast available literature estimates of K_a, K_h, and h_m including specialized literature in environmental engineering, chemical engineering, and physical and atmospheric chemistry. This approach was effective in comparing variation between parameter estimates.

Dissociation and Henry's Law Constant

Pure Water-Ammonium Solutions

Reported estimates of K_a for pure water-NH_4^+ solutions are plotted in figure 1 for 0°C to 35°C. The figure shows relatively small variation among reported values, particularly at temperatures less than 30°C. Data sets that

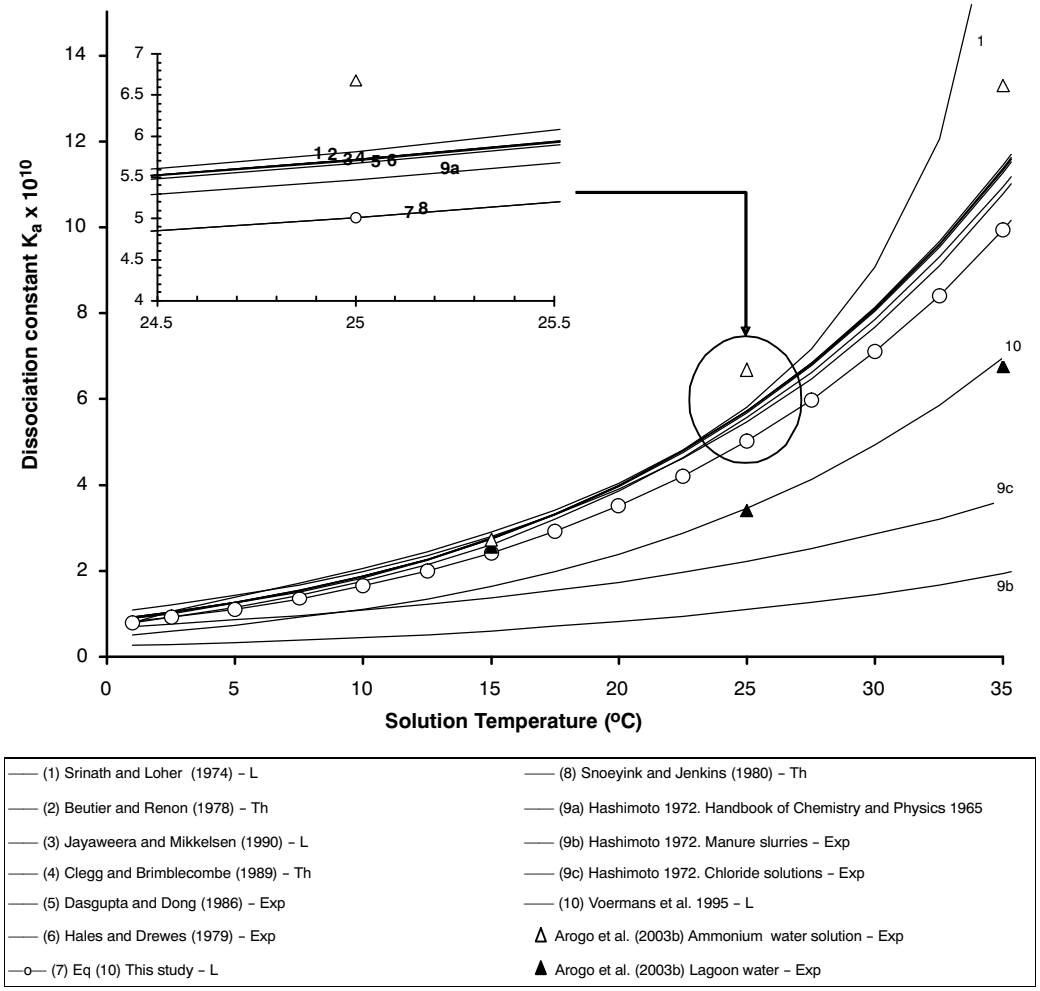


Figure 1. Ammonia dissociation constant in pure water with additional data for the dissociation constant in ammonium chloride and manure solutions (Exp = experimental measurements, Th = thermodynamic models, and L = literature values).

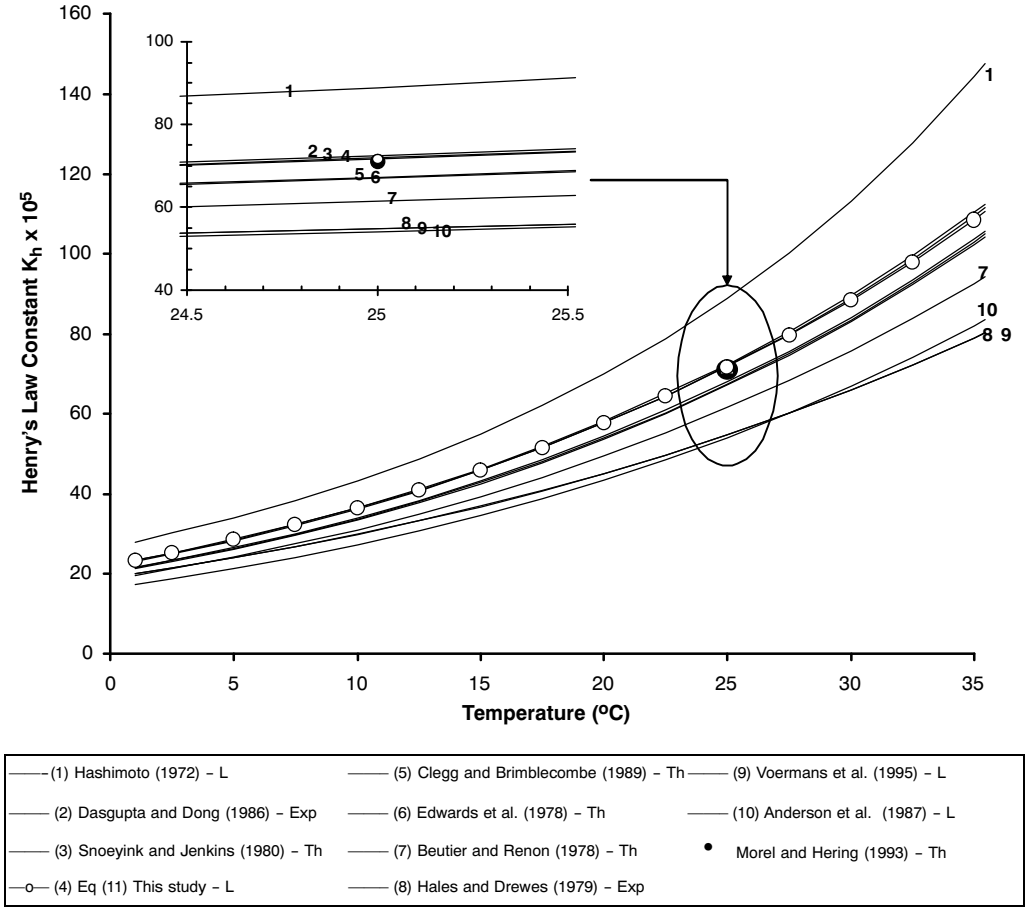


Figure 2. Henry's law constant for NH_3 in diluted water solutions and standard values at 25°C (Exp = experimental measurements, Th = thermodynamic models, and L = literature values).

deviate greatly from the majority are those from manure or solutions containing materials in addition to NH_4^+ and water.

Experimental measurements of K_a are difficult to obtain, requiring simultaneous measurements of the concentrations of NH_3 and NH_4^+ in solution, temperature, and pH. With the advent of sophisticated thermodynamic models and increased computer power, thermodynamic calculations have provided more accurate estimations. The data presented suggest that experimental measurements (data sets 5, 6, and Δ in fig. 1; data sets 2 and 8 in fig. 2) compare well to thermodynamic model based estimates (data sets 2, 4, and 8 in fig. 1; data sets 3, 5, 6, and 7 in fig. 2).

Ammonia in solution is not only in equilibrium with NH_4^+ but also with NH_3 at the water-air interface. Henry's law quantifies the equilibrium of NH_3 in solution with the gaseous NH_3 in the air immediately above the solution. The quantitative definition of the Henry's law constant varies greatly in the literature. Staudinger and Roberts (1996), Ni (1999), and Renard et al. (2004) have pointed out variations in the units used to quantify the value of K_h, as well as in the values obtained when the estimates are transformed to common units. Ni (1999) recommended the use of a dimensionless expression, but as was described by Staudinger and Roberts (1996), there are at least two alternative expressions in a dimensionless form. We chose to use the dimensionless form based upon the equilibrium concentrations expressed in equations 2 and 3. Temperature is the primary factor affecting the Henry's law constant in pure water. Figure 2 is a plot of literature K_h estimates transformed to the common units of equation 3.

Figures 1 and 2 illustrate the need for standard relationships that correctly represent NH_4^+ dissociation, reduce variability, and allow comparisons between volatilization models. From this summary, we found that thermodynamic estimates based on first principles appropriately described the behavior of K_a and K_h. Simple equations for application under the conditions of atmospheric pressure and temperature between 0°C and 40°C were derived using the van't Hoff equation (a general relationship representing temperature effects on equilibrium constants), as suggested by Snoeyink and Jenkins (1980) and Stumm and Morgan (1996). This general equation was parameterized using K_a and K_h data for standard temperature and pressure (25°C, 1 atm) and the assumption of constant enthalpy of reaction for the temperature range from 0°C to 40°C (Snoeyink and Jenkins, 1980; Stumm and Morgan, 1996). The following expressions were obtained:

$$\log(K_a) = -9.3 - \left\{ \frac{52.21}{\ln(10)R} \left(\frac{1}{T_2} - \frac{1}{298.15} \right) \right\} \quad (8)$$

$$\log(K_h) = \log\left(7.17 \cdot 10^{-4} \cdot \frac{298.15}{T} \right) - \left\{ \frac{34.19}{\ln(10)R} \left(\frac{1}{T} - \frac{1}{298.15} \right) \right\} \quad (9)$$

where R is the universal gas constant (0.008134 kJ mol^{-1} K^{-1}). For easier use, equations 8 and 9 can be simplified to:

$$K_a = 10^{0.05 - 2788/T} \quad (10)$$

$$K_h = (0.2138/T)\, 10^{6.123 - 1825/T} \quad (11)$$

The equation obtained for K_h includes a transformation from mole fraction to (g NH_3-N m^{-3} air) / (g NH_3-N m^{-3} liquid) by applying the ideal gas law, as suggested by Staudinger and Roberts (1996). Equations 10 and 11 are plotted in figures 1 and 2 using a line through open circles. These equations are relatively simple and represent well the behavior of K_a and K_h in the temperature range most common to agricultural systems. Therefore, they are our recommended models for determining K_a and K_h for pure water solutions of NH_4^+ under atmospheric pressure and temperatures from 0°C to 40°C. When it is necessary to estimate K_a and K_h for a broader range of temperature, more sophisticated equations can be used, such as those provided by Clegg and Brimblecombe (1989), Hales and Drewes (1979), or Beutier and Renon (1978).

Manure Ammoniacal Nitrogen System

Differences in K_a and K_h between NH_4^+-water solutions and manure have been documented (Arogo et al., 2003b; Hashimoto and Ludington, 1971; Zhang, 1992). These differences are attributed to two factors: (1) the presence of organic matter and (2) the presence of other ions in solution.

Organic Matter in Manure

Evidence of NH_3 and NH_4^+ adsorption to soils and organic matter has been well documented in the literature (Boatman and Murray, 1982; Mackin and Aller, 1984; McNevin et al., 1999; Nielsen, 1996; Nommik and Vahtras, 1982; Staudinger and Roberts, 1996). In extrapolating these results to manure, it has been assumed that the organic matter in manures will adsorb some of the NH_4^+ and NH_3 in solution. This would result in reduced chemical activity of these compounds without any apparent reduction in the analytical values of total ammoniacal N in the manure. This has been suggested as a possible cause for the difference in measured K_a values between NH_4^+-water solutions and manure (Arogo et al., 2003b; De Visscher et al., 2001; Hashimoto and Ludington, 1971; Liang et al., 2002; Zhang, 1992).

However, the organic matter present in fresh manure is composed primarily of partially degraded feed and digestion residuals. The structure of these materials is quite different from the more highly degraded, stable, and complex structure of the humic substances in soil organic matter. Therefore, a low potential for NH_3 adsorption to the organic matter in fresh manure may exist compared to the adsorption potential in soils, humic acids, or peat (Nommik and Vahtras, 1982; Schwarzenbach et al., 2003; Stumm, 1992).

Another factor that reduces the NH_4^+ adsorption to soils and organic matter is the competition for available adsorption sites and the displacement of adsorbed NH_4^+ by other ions present. Figure 3 shows the mineral analysis of a typical fresh dairy manure sample taken from the Pennsylvania State University dairy farm together with manure compositions found in the literature. Large amounts of calcium, magnesium, and potassium are common in manure, although not all in soluble reactive form, since a fraction is precipitated in minerals or associated with other ions in solution. Nevertheless, there are calcium, magnesium, and potassium ions in solution that compete with NH_4^+ for adsorption sites in the manure organic matter.

From studies of NH_4^+ adsorption in soils, it has been observed that different cations are adsorbed by organic matter with different intensities. According to the lyotropic series (ease to be removed from soil colloids), the intensity is greater for calcium and magnesium, followed by potassium and NH_4^+, with sodium having the lowest adsorption intensity (Bohn et al., 1985). The chance for NH_4^+ adsorbing to the organic matter in manure is further diminished by the presence of these cations. Therefore, typical fresh manure with less than 12% volatile solids composed of partially degraded feed materials and many cations competing for adsorption sites suggests low NH_4^+ adsorption to the organic matter present in the manure (*ASAE Standards*, 2003). One of the few studies reporting NH_4^+ adsorption in manure

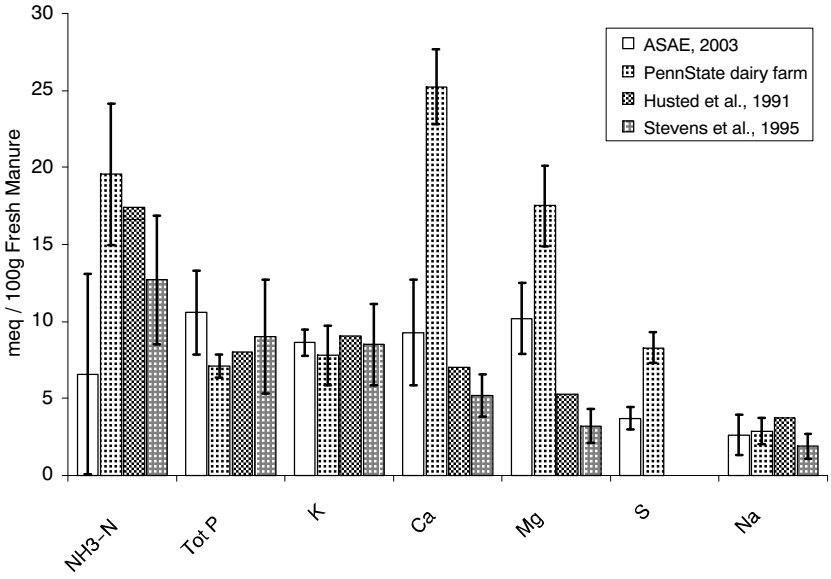


Figure 3. Important ions found in fresh manure.

organic matter found that less than 10% of the total ammoniacal N was adsorbed to the organic matter in cattle manure digesters where the solution had a total solids content of about 9% (Hafner et al., 2006a, 2006b).

Interaction with Other Ions in Manure

The interaction of NH_3 and NH_4^+ with other ions in manure solutions has been discussed in the literature, although few quantitative determinations have been made (Arogo et al., 2003b; Hafner et al., 2006a). Arogo et al. (2003b) found that the value of the dissociation constant for NH_3 in anaerobic lagoon liquid (0.3% to 0.5% solids) was 0.94 to 0.5 of the value measured in de-ionized water. They found no explanation for the difference, but indicated that some portion could be explained by the effects of other ions in the solution and ionic strength. Génermont and Cellier (1997) estimated that up to 25% of the overestimation of NH_3 volatilization in their study could have been due to not using ionic strength corrections. Liang et al. (2002) found the value of the dissociation constant for NH_3 in anaerobic swine lagoon liquid with 0.25% solids to be half the value obtained for de-ionized water.

The effect of other ions in solution on the equilibrium of NH_3 and NH_4^+ can be incorporated into equation 4 using activity coefficient corrections. Equation 4 assumes that NH_3 is dissolved in pure water. In solutions where other ions are present, such as manure or other mixed electrolyte solutions, interaction with the other ions affects the chemical activity of NH_3 and NH_4^+. Therefore, the chemical activity can no longer be assumed to be equal to the analytical concentration.

The activity of NH_3 and NH_4^+ can be estimated from the concentrations of these species in solution multiplied by their corresponding activity coefficients $\gamma_{NH_{3(aq)}}$ and $\gamma_{NH_4^+}$ (Morel and Hering, 1993; Snoeyink and Jenkins, 1980):

$$K_a = \frac{[NH_3]\gamma_{NH_3(aq)}[H^+]\gamma_{H^+}}{[NH_4^+]\gamma_{NH_4^+}} \tag{12}$$

From equation 12, a corrected dissociation constant K_a^c can be obtained to adjust the dissociation constant of NH_3 in pure water for interactions with other ions in mixed electrolyte solutions:

$$K_a^c = K_a \frac{\gamma_{NH_4^+}}{\gamma_{NH_3(aq)}\ \gamma_{H^+}} \tag{13}$$

Likewise for the Henry's law constant:

$$K_h^c = \frac{[NH_3]_{gas}}{[NH_3]_{solution}\gamma_{NH_3(aq)}} = \frac{K_h}{\gamma_{NH_3(aq)}} \tag{14}$$

Hafner et al. (2006a, 2006b) experimentally determined the activity of NH_3 in solution with cattle manure. They used electrode measurements, the extended Debye-Huckel model, and the Pitzer model for electrolyte solutions to estimate the activity of NH_3 in cattle manure digesters with 10% to 13% total solids. Their result suggests that the activity of NH_3-N in manure digesters could be overpredicted by 45% to 200% using current mass balance and equilibrium methods without activity corrections. They found good agreement between measured NH_3 activity in manure digester solutions and that predicted by the extended Debye-Huckel and Pitzer models.

Activity coefficients can be determined using the ionic strength of manure estimated using electrical conductivity measurements, as suggested by Snoeyink and Jenkins (1980):

$$I = 1.6 \cdot 10^{-2}\ (EC) \tag{15}$$

where EC is the electrical conductivity (mmhos cm^{-1}). Measured electrical conductivity of manure is usually less than 30 mmhos cm^{-1}, and ionic strength is less than 0.5 mol L^{-1}. Within this range of ionic strength, the activity coefficient of NH_4^+ can be estimated using the Davies approximation equation (Morel and Hering, 1993; Stumm and Morgan, 1996):

$$\log \gamma_{NH_{4(aq)}^+} = -0.5\ (c^2)\left(\frac{\sqrt{I}}{1+\sqrt{I}} - 0.3I\right) \tag{16}$$

where c is the charge of the ion (c = 1 for NH_4^+). Morel and Hering (1993) suggest that for neutral molecules, log γ is approximately equal to $0.1 \times I$. This empirical relation can be used to estimate the activity coefficient of NH_3 in solution when other ions are present. Direct measurement of manure or buffered solution pH includes an ion activity correction; therefore, no correction for H^+ is necessary in equation 13. When the ionic strength of manure or other electrolyte solution exceeds 0.5 mol L^{-1}, the Davies equation is no longer appropriate. A more complex ion interaction model such as the Pitzer equations for electrolyte solutions must be used (Hafner et al., 2006a, 2006b; Parkhurst and Appelo, 1999; Pitzer, 1991).

To further explore the effects of ionic strength on the Henry's law and NH_3 dissociation constants, a water solution with ion concentrations similar to that of the dairy manure illustrated in figure 3 was evaluated through a range in ionic strength. The effects of increasing ionic strength on the Henry's law and NH_3 dissociation constants were calculated using equations 13 and 14 rearranged to determine the ratio of the corrected constants, K_a^c or K_h^c, over the uncorrected constants K_a or K_h. As described above, the activity of NH_3 was calculated using the procedure of Morel and Hering (1993) or the modeling software PHREEQC V.2 (Parkhurst and Appelo, 1999). The activity of NH_4^+ was calculated using either PHREEQC or the Davies equation. The simulation results show that the effect of other ions commonly present in manure, even at high ionic strengths, could only explain up to a 50% reduction in the NH_3 dissociation constant of manures when compared to pure NH_4^+-water solutions (fig. 4).

MASS TRANSFER COEFFICIENT

Most mass transfer models used to predict NH_3 volatilization are based on the two-film interface transport model developed by Lewis and Whitman (1924). The model assumes the formation of hypothetical thin films on both sides of the interface, where the transported substance moves through these liquid and gaseous quiescent films by diffusion. For highly soluble gases, the resistance offered by the gas film is very restrictive compared to a negligible resistance offered by the liquid film. This is the case for NH_3 transport through a water-air interface (Schwarzenbach et al., 2003).

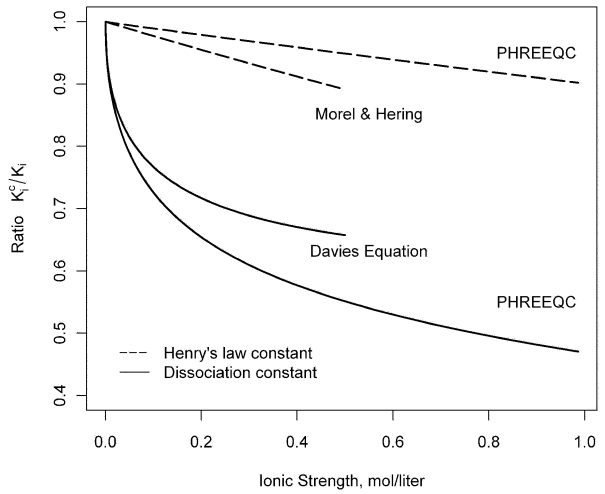


Figure 4. Effect of ionic strength on the dissociation and Henry's law constants shown as the ratio K_a^c / K_a or K_h^c / K_h.

Previous Models

Mass transfer coefficients reported for NH_3 volatilization vary widely. Part of the variation is due to the physical characteristics of the mass transport process, i.e., air velocity and laminar versus turbulent flow. The major source of variation, though, appears to be the cumulative effect of assuming values for the Henry's law and dissociation constants in the development of empirical mass transfer models. Most frequently, values for K_a and K_h are obtained from literature sources, and the value of h_m is fitted to experimental data. As illustrated in figures 1 and 2, variation in assumed values for K_a and K_h can be large, therefore increasing the variability of h_m. Various published mass transfer coefficients are compared in figure 5 using common units of g NH_3-N $m^{-2}s^{-1}$. For this comparison, all coefficients were determined for a temperature of 20°C (unless noted otherwise), pressure of 1 atmosphere, and a standard concentration gradient across the interface of 1 g NH_3-N m^{-3} air. Henry's law together with the universal gas law were used to obtain an equivalent concentration gradient of 1840 g NH_3-N m^{-3} liquid, and this gradient was used when the mass transfer coefficient was reported in terms of liquid phase concentration units. Using a standard concentration gradient of one unit across the interface isolated the effect of chemical equilibrium and allowed for an effective comparison. Figure 5 illustrates the wide variation in reported coefficients.

As reported by Ni (1999), the most common model used to predict NH_3 volatilization in agricultural systems is that derived by Haslam et al. (1924). They measured the mass transfer coefficient by blowing humidified NH_3 gas up through a 76 mm vertical glass tube with a thin film of distilled water traveling down the tube wall. The empirical equation obtained is specific to the mass transfer system used in their experimental setup, as is clearly stated in their results. Their expression, though, has been used in many models to provide a relatively good representation of volatilization from manure surfaces (Elzing and Monteny, 1997b; Muck and Steenhuis, 1982; Koelliker and Miner, 1973; Anderson et al., 1987). Nevertheless, estimates of the mass transfer coefficient based on this expression (data sets 1, 1a, 1b, and 2 in fig. 5) are grouped well above every other relationship from the reviewed literature. The empirical model reported by Zhang et al. (1994) gave similar predictions. Their model was obtained using swine manure in wind tunnel experiments over a narrow air velocity range. The dashed line for data set 3 in figure 5 shows the extrapolation of the Zhang et al.

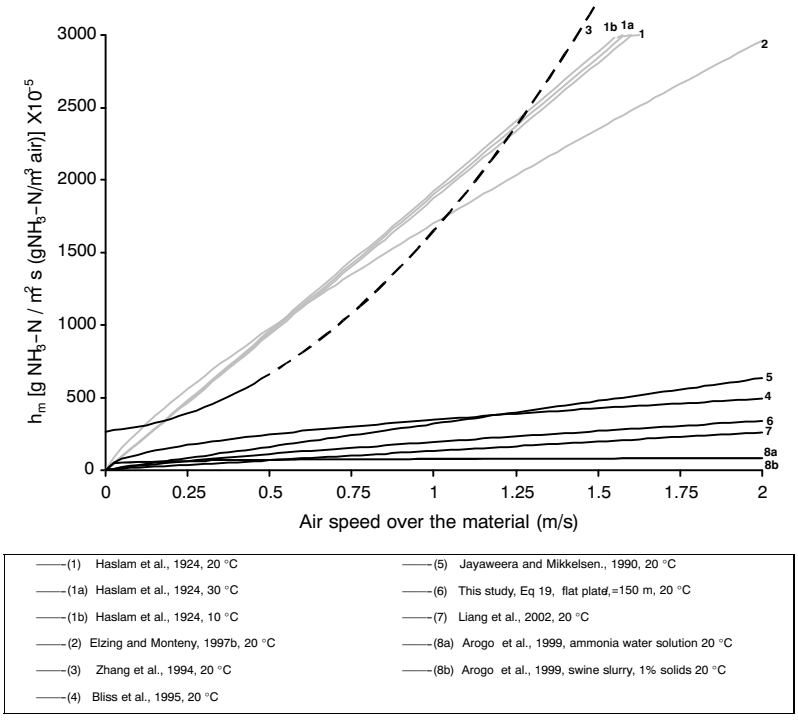


Figure 5. Mass transfer coefficient relationships reported in the literature with values expressed as flux (g NH_3-N s^{-1} m^{-2} surface) obtained when constant temperature (20°C or as indicated in the legend) and atmospheric pressure (1 atm) are used over a concentration gradient of 1 g NH_3-N m^{-3} air or 1840 g NH_3-N m^{-3} liquid.

(1994) model above the range of air velocity used in their experiments.

Recommended Model

The mass transport process is primarily a function of the density, diffusivity, and viscosity of the substance being transported and the transporting medium. In this case, the substance transported is NH_3 and the transporting medium is moist air. In practice, the amount of NH_3 volatilized is small, and thus does not affect the physical characteristics of the air that it is entering. Therefore, the most important molecular properties affecting the transport process are the diffusivity of NH_3 in air ($D_{NH3\text{-}air}$), air viscosity (μ_{air}), and air density (ρ_{air}). The combined effects of these parameters on the mass transfer process can be modeled using dimensional analysis to provide the following relationship (Arogo et al., 1999; Incropera et al., 1996; Perry et al., 1997; Schwarzenbach et al., 2003):

$$\mathrm{Sh} = C\,\mathrm{Re}^{n}\,\mathrm{Sc}^{m} \tag{17}$$

with

$$\mathrm{Sh} = \frac{h_m l}{D_{NH_3-Air}}$$

$$\mathrm{Re} = \frac{lU\rho_{air}}{\mu_{air}}$$

$$\mathrm{Sc} = \frac{\mu_{air}}{\rho_{Air} D_{NH_3-Air}}$$

where Sh is the Sherwood number, Re is the Reynolds number, Sc is the Schmidt number, h_m is the mass transfer coefficient (m s^{-1}), l is the characteristic length determining airflow (m), U is the average air velocity (m s^{-1}), and C, n, and m are parameters that depend on the characteristics of the flow and are determined experimentally.

In agricultural systems, NH_3 volatilization primarily occurs from a relatively flat surface such as a barn floor, manure storage surface, or land-applied manure. The theory for mass transfer through a boundary layer over a flat surface is well documented for both laminar and turbulent airflow conditions (Incropera et al., 1996; Schlichting and Gersten, 2000). For the conditions prevalent in agricultural systems, turbulent flow can be assumed as a result of buoyancy created by spatial thermal differences, rough surfaces, and the presence of obstacles surrounding the volatilization surface (Rosenberg et al., 1983). Since turbulence can be expected under most conditions, a relationship for average mass transfer with turbulent flow over a flat plate was selected for our model (Incropera et al., 1996; Perry et al., 1997):

$$\mathrm{Sh} = 0.0365\,\mathrm{Re}_L^{0.8}\,\mathrm{Sc}^{0.33} \tag{18}$$

Relationships are available for calculating D_{NH_3-Air} as a function of temperature, pressure, and the molecular characteristics of the compounds (Perry et al., 1997); μ_{air} as a function of temperature; and ρ_{air} as a function of temperature, pressure, and humidity (Arogo et al., 1999). Incorporating relationships for D_{NH_3-Air}, μ_{air}, and ρ_{air} with equation 18, the h_m for NH_3 transfer in air can be determined as a function of temperature, air velocity, and the characteristic length of the emitting surface:

$$h_m = 0.000612\,U^{0.8}\,T^{0.382}\,l^{-0.2} \tag{19}$$

To simplify the equation, dry air and standard atmospheric pressure were assumed.

Experimental Verification

The fundamental relationships developed here were used to predict NH_3 volatilization rates from dairy cattle manure and buffered NH_4^+-water solutions measured under controlled laboratory conditions in a companion study by Chaoui et al. (2009). The manure samples, a mixture of feces and urine collected from the floor of a free-stall barn, were kept under controlled laboratory conditions for 12 h prior to the trials to allow the manure to reach the desired temperature and for the urea to be decomposed into ammoniacal N. Ammonium solutions were made with laboratory-grade chemicals consisting of a 0.1 mol L^{-1} sodium phosphate buffer and 1 to 5 g NH_4^+-N added as ammonium nitrate. The pH of solutions was adjusted to between 7.0 and 8.0 using sodium hydroxide. Average electrical conductivities were 23 and 50 mmhos cm^{-1} for the manure and buffered NH_4^+ solution, respectively. Ammonia volatilization at different air velocities and temperatures was measured using a flux chamber technique developed by Wheeler et al. (2007).

Electrical conductivity measurements from manure and ammonium solutions were used to estimate ionic strength (eq. 15). The ionic strength was then used to estimate the activity coefficient of NH_4^+ and NH_3 (eq. 16). Ionic strengths of manure were within the range suitable for using the Davies equation (I < 0.5 mol L^{-1}). The ionic strength of the buffered NH_4^+ solutions exceeded this range, so their activity was estimated using the Pitzer equations for electrolyte solutions as implemented in the USGS geochemical modeling software PHREEQC V.2 (Parkhurst and Appelo, 1999). With these calculated activity coefficients, equations 10, 11, 13, and 14 were used to determine the equilibrium concentration of NH_3 in the air immediately above the surface of the manure or buffered NH_4^+ solution.

Using the air velocity and temperature of each experimental trial, the mass transfer coefficient was determined using equation 19. The characteristic length l was set to the length of the manure surface in the flux chamber. The roughness of the manure surface, turbulent flow in the recirculation pipe of the flux chamber, and the presence of screens for flow distribution purposes made it reasonable to assume turbulent conditions. Measurements of the velocity profile inside the flux chamber also showed a flat velocity profile similar to that expected for turbulent flow in rectangular pipes (Wheeler et al., 2007; Schlichting and Gersten, 2000).

Ammonia volatilization rates from manure and ammonium buffered solution measured by Chaoui et al. (2009) were compared to emission rates predicted by our recommended relationships and those predicted by previous NH_3 volatilization models proposed by Muck and Steenhuis (1982), Monteny et al. (1998), Cortus et al. (2008), Teye and Hautala (2008), and Zhang et al. (1994). Our relationships predicted reasonable NH_3 emissions for both manure and buffered NH_4^+ solution (figs. 6 and 7). At higher

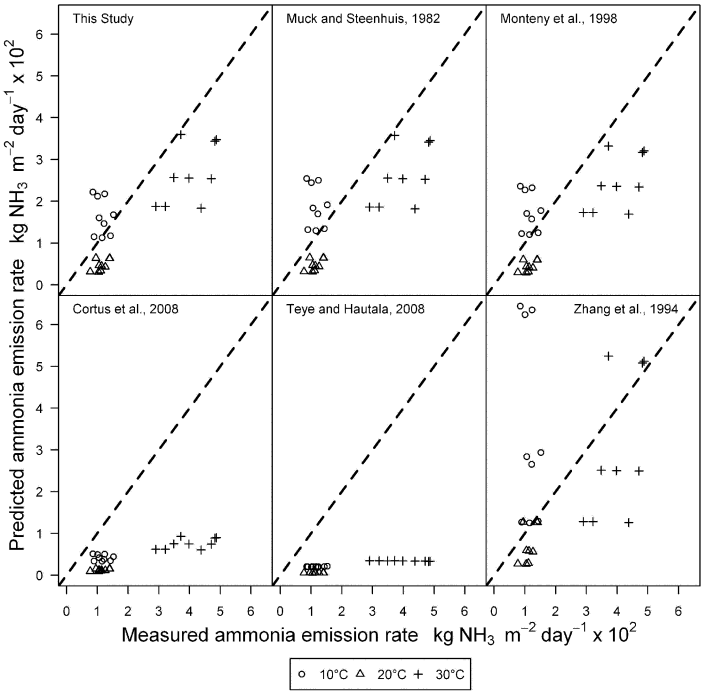


Figure 6. Ammonia emission rates from dairy cattle manure measured by Chaoui et al. (2008) compared to predictions for the same experimental conditions using the methodology suggested in this research and by similar expressions found in ammonia volatilization models proposed by Muck and Steenhuis (1982), Monteny et al. (1998), Cortus et al. (2008), Teye and Hautala (2008), and Zhang et al. (1994).

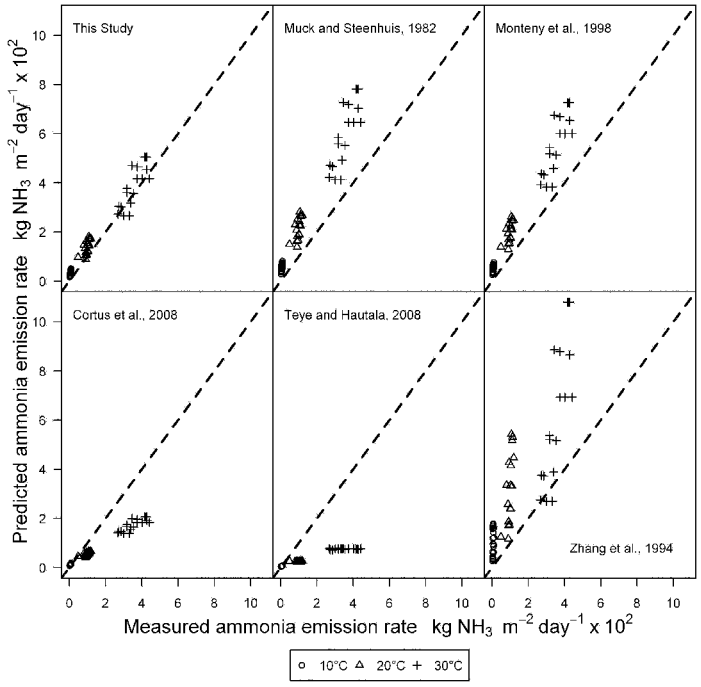


Figure 7. Ammonia emission rates from buffered ammonium solution measured by Chaoui et al. (2008) compared to predictions for the same experimental conditions using the methodology suggested in this research and by similar expressions found in ammonia volatilization models proposed by Muck and Steenhuis (1982), Monteny et al. (1998), Cortus et al. (2008), Teye and Hautala (2008), and Zhang et al. (1994).

temperatures, measured emissions from manure were greater than that predicted by all models. A cause for this cannot be documented, but it may be due to changes in pH occurring on the manure surface. Laboratory measurements have shown that the surface pH can increase rapidly upon exposure to air, and this change will increase NH_3 volatilization (Chaoui et al., 2009). As reflected in equation 4, a very small change in pH can have a substantial effect on dissociation and the resulting emission.

Predictions made using the relationships of Muck and Steenhuis (1982) and Monteny et al. (1998) are in good agreement with emission rates measured in manure, but emission rates from ammonium-buffered solutions were overpredicted as compared to measured values. For both of these studies, the dissociation constant of ammonium in pure water was adjusted by 1/6 as reported by Hashimoto and Ludington (1971) together with the mass transfer coefficient of Haslam et al. (1924). Predictions made by the relationships found in the models of Cortus et al. (2008) and Teye and Hautala (2008) used the relationship proposed by Monteny et al. (1998) for the dissociation constant, but did not use the corresponding relationship for the mass transfer coefficient. The mass transfer relationship used by Cortus et al. (2008) was based on laminar boundary layer flow, and that used by Teye and Hautala (2008) was based on turbulent boundary layer flow. Using the relationships for the dissociation constant reported by Hashimoto and Ludington (1971) separate from the relationship for the mass transfer coefficient reported by Haslam et al. (1924) resulted in disagreement with the measured emissions from manure or buffered NH_4^+ solution. The relationship proposed by Zhang et al. (1994) used the dissociation constant of NH_4^+ in pure water adjusted by 1/5, together with an empirically determined mass transfer coefficient. Their relationship predicted the trend of emission rates measured by Chaoui et al. (2009) in manures but overpredicted the emission rates measured from the buffered solution.

DISCUSSION

Independent theoretical relationships were used to develop the proposed emission model. This mechanistic, process-based NH_3 volatilization model successfully predicted NH_3 volatilization rates from dairy cattle manure and buffered NH_4^+ solutions under closely controlled laboratory conditions. An important improvement over previous models was the use of electric conductivity measurements to estimate the ionic strength and activity coefficients of NH_3 and NH_4^+ and the use of these values to adjust the dissociation and Henry's law constants.

Reasonable predictions have also been obtained using the equations for NH_3 dissociation proposed by Hashimoto (1972) along with the mass transfer coefficient proposed by Haslam et al. (1924) (Chaoui et al., 2008). This previous model used 1/6 of the dissociation constant K_a in pure water as the dissociation constant in manure along with a mass transfer coefficient higher than others reported in the literature (fig. 5). It appears that compensating differences of Hashimotos's (1972) dissociation constant and the mass transfer coefficient of Haslam et al. (1924) together result in reasonable predictions of NH_3 emissions. Use of the 1/6 factor to correct the NH_3 dissociation constant in manure appears to underestimate emissions from manure but overestimate those for buffered NH_4^+ solution (Chaoui et al., 2008). Theoretical justification was not found for adjusting the dissociation constant of NH_4^+ in manures to 1/6 that in pure water. It is unlikely that the organic matter present in the manure is responsible, and the interaction with other ions, even at extreme ionic conditions, results in ratios of only 1/2 to 2/3.

The relationships presented satisfactorily predicted NH_3 volatilization under controlled laboratory conditions, but extrapolation to barns and manure storages under farm conditions is still required. More information and ultimately a model are needed to predict the change in pH on exposed manure surfaces (Chaoui et al., 2008). A method is also needed to determine characteristic length parameters for various manure surfaces. In situations such as large manure storages or manure applied over field surfaces, the characteristic length parameter is not obvious or easily determined. For these conditions, aerodynamic or micrometeorological volatilization estimation methods based on a fully developed turbulent boundary layer and logarithmic wind profile can be applied. These methods are based on the same principles as equation 19, but the main parameters determining the mass transfer are those characterizing the turbulent flow profile (Génermont and Cellier, 1997; Jayaweera and Mikkelsen, 1990; Olesen and Sommer, 1993; Ro and Hunt, 2007; Rosenberg et al., 1983; Shah et al., 2006). Determination of mass transfer coefficients for NH_3 volatilization from barn floors is especially challenging. The assumption of uniform turbulent flow and velocity profile is not applicable, and a characteristic length for use with equation 19 is not well defined (Harper, 2005). Applying this methodology to farm conditions requires the generation and compilation of further data, but will provide a more robust NH_3 volatilization model.

Mass transfer processes in mechanically and naturally ventilated structures depend on the air velocity near the volatilization surface and the turbulence of the airflow. Velocity and flow characteristics in naturally ventilated barns are both influenced by building geometry, size and distribution of ventilation inlets and outlets, heat production, local wind conditions, obstacles inside the buildings (pens, stall walls, doors, animals), circulation fans, and animal activity (Demmers et al., 2000; Gebremedhin and Wu, 2005; Morsing et al., 2000; Ogilvie et al., 1990; Strøm et al., 2002). Complex flows and air velocity patterns inside naturally ventilated buildings have been observed in wind tunnel scaled models and with computational fluid dynamic simulations (Demmers et al., 2000; Gebremedhin and Wu, 2005; Morsing et al., 2000). Simpler models that make reasonable predictions of near-floor velocity based on inlet configuration, building geometry, and meteorological available wind velocity data will further improve process-based models for agricultural systems.

The mass transfer models presented are based on the presence of a rigid or motionless surface at the intersection of the flowing air and volatilizing surface (Schwarzenbach et al., 2003). This can be assumed for manure with a high viscosity or surface crust. In the case of liquid surfaces such as lagoons, this assumption is not always valid. For low air velocities, the liquid behaves as a smooth rigid surface. When the air velocity reaches 3.7 to 5 m s^{-1}, ripples and waves form, and wind-induced turbulence in the liquid changes the dynamics of the mass transfer process (Crusius and Wanninkhof, 2003; Deacon, 1977; Ro and Hunt, 2007; Schwarzenbach et al., 2003;). Wind-induced turbulence reduces the mass transfer resistance in the liquid interface, which is especially important for low-solubility gases like oxygen; its effect on soluble gases such as NH_3 is not as well understood.

CONCLUSIONS

Ammonia volatilization models rely heavily on three parameters: the dissociation constant (K_a), the Henry's law constant (K_h), and the mass transfer coefficient (h_m). From an extensive literature review, relationships that provide the best estimates of parameters K_a and K_h in pure aqueous solutions were identified. These estimates can be applied to manure or other aqueous NH_4^+ solutions using electrical conductivity measurements to estimate appropriate activity coefficient corrections for the interaction with other ions in solution. Using our relationships for K_a and K_h together with activity corrections improved the prediction of NH_3 dissociation from dairy cattle manure under controlled laboratory conditions.

Adjusting the value of K_a for ammonia in pure water solution by a factor of 1/6 to obtain the value of K_a for manures closely predicted the experimental results only when used together with the mass transfer coefficient relationship reported by Haslam et al. (1924) or when the mass transfer coefficient was obtained by fitting experimental data, as was the case in the study by Zhang et al. (1994). It is unlikely that the organic matter present in manure or interaction with other ions will result in K_a values that are about 1/6 of the values of K_a in pure water solutions.

A theoretically derived mass transfer coefficient for turbulent flow over a flat plate was successfully used to predict measured NH_3 volatilization rates under laboratory conditions, but only when used with the recommended relationships for predicting K_a and K_h with appropriate activity coefficient corrections estimated from electrical conductivity measurements. Validation of the proposed relationships for animal facilities is still necessary in order for them to be effectively used in ammonia volatilization prediction models.

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